



PR-DETR: Injecting position and relation prior for dense video captioning

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ABSTRACT

Dense video captioning is a challenging task that aims to localize and caption multiple events in an untrimmed video. Recent studies mainly follow the transformer-based architecture to jointly perform the two sub-tasks, *i.e.*, event localization and caption generation, in an end-to-end manner. Based on the general philosophy of detection transformer, these methods implicitly learn the event locations and event semantics, which requires a large amount of training data and limits the model's performance in practice. In this paper, we propose a novel dense video captioning framework, named PR-DETR, which injects the explicit position and relation prior into the detection transformer to improve the localization accuracy and caption quality, simultaneously. On the one hand, we first generate a set of position-anchored queries to provide the scene-specific position and semantics of potential events as position prior, which serves as the initial event search regions to eliminate the implausible event proposals. On the other hand, we further design an event relation encoder to explicitly calculate the relationships between event boundaries and encode them into the relation mask as relation prior to improve the coherence of the captions. Extensive ablation studies are conducted to verify the effectiveness of the position and relation prior. Experimental results also show the competitive performance of our method on ActivityNet Captions and YouCook2 datasets.

1. Introduction

As a vital branch of video understanding, video captioning [1–4] aims to generate a single caption for a short video clip. To handle the long and untrimmed videos in reality, dense video captioning [5–8] is proposed to jointly localize multiple events in the video and describe each event with natural language. Due to its great potential in practical applications, such as video retrieval [9,10] and multi-modal analysis [11,12], dense video captioning has been widely studied since its emergence. Compared with conventional video captioning, the requirement for accurate multi-event localization and fine-grained captioning makes dense video captioning more difficult in algorithm development, and more practical in engineering deployment.

Early dense video captioning methods [13,14] mainly follow a two-stage paradigm, first localizing events in the video, and then generating corresponding captions. In recent years, with the success of DETection TRansformer (DETR) [15] in object detection, several works [7,16,17] adopt the DETR-like architecture to perform event localization and caption generation simultaneously. For example, PDVC [7] extends DETR into the dense video captioning task, which considers dense video captioning as a set prediction task to predict the locations and captions of multiple events in parallel. CM2 [16] enhances the DETR-like model with cross-modal retrieval from external memory.

Despite the significant advances achieved by these methods, there are still two critical challenges in dense video captioning that limit their further improvement. Firstly, the distribution pattern of events is complex and varies significantly between different scenarios. As shown in Fig. 1(a), the location of multiple events tends to exhibit a hierarchical distribution in some scenarios, where temporal inclusion and overlap occur between different events. While in some other scenarios, multiple events tend to be relatively short and follow a step-by-step distribution. This diverse distribution of events makes event localization more difficult. Secondly, event descriptions within a single video usually exhibit logical dependencies and semantic co-occurrence, especially for events that are temporally connected. As shown in Fig. 1(b), event 2 and event 3 in the first scenario include co-occurring semantics such as 'balance beam routine'. In the second scenario, events 3 and 4 collectively describe a complete process of washing a sink. This poses a significant challenge for the model to maintain the coherence of the generated captions, constraining the captioning performance of the model.

To handle the above challenges, previous methods adopt a set of learnable embeddings as event queries to predict the event locations and model the event relations through implicit self-attention between

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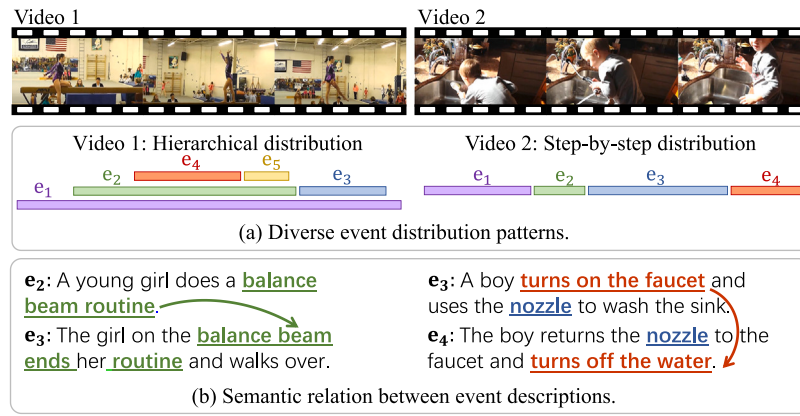


Fig. 1. Illustration of the event locations and captions in different scenarios within the ActivityNet dataset. (a) The locations of multiple events follow diverse distribution patterns across scenarios. In some scenarios, the events follow a hierarchical distribution and vary widely in length with overlap and inclusion, while in some other scenarios, the events exhibit a step-by-step distribution. (b) There exist logical dependencies and semantic co-occurrence between event captions within a single video, making it challenging to maintain the coherence of event captions.

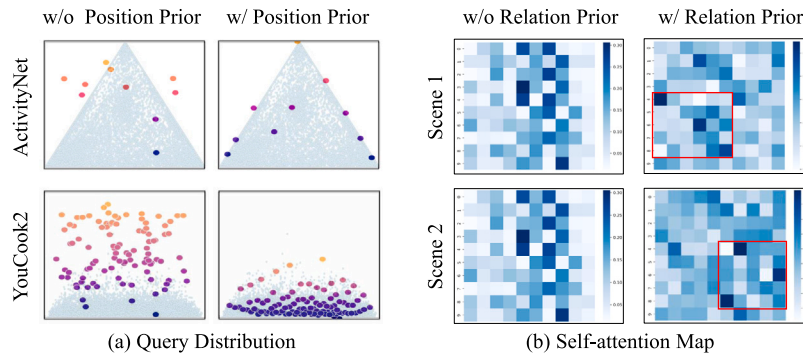


Fig. 2. Visualization of the Query Distribution across different datasets and the Self-attention Map across different scenarios with and without the Relation Prior. (a) Visualization of query distribution with and without position prior. x-axis represents the normalized event center coordinates, and the y-axis denotes the normalized event duration. Compared to previous methods, adding the Position Prior significantly brings the learned query distribution closer to the actual distribution in the dataset. (b) Visualization of the self-attention map between event queries with and without relation prior. For different scenarios, previous methods show the same attention map. With the relation prior, the self-attention map better reflects the semantic correlations between potential events.

queries. This implicit learning strategy is insufficient for capturing structured event distributions and maintaining coherence across event captions, which may bring ambiguity to the model. As shown in Fig. 2(a), the learned queries are distributed across the entire event space after training, and some of them represent implausible event locations. In the DETR-like architecture, the self-attention map represents the relationship between event queries. As shown in Fig. 2(b), the model has the same attention map for different scenarios, which inadequately reflects the relationship between events. This motivates us to explore the event distribution pattern as the position prior to provide more reliable event search regions. We also model explicit event relations as the relation prior to guide the event interactions in the decoding process, thus improving the localization accuracy and caption quality of the DETR-like dense video captioning models.

In this paper, we propose a novel framework, named PR-DETR, to inject the Position and Relation prior into the DETR-like architecture for the dense video captioning task. For the position prior, we generate a set of position-anchored queries that contain the initial search regions of event location and semantics from the data distribution. We first initialize the query with the clustering centers of the ground-truth event location. After feature aggregation, the query provides scene-specific search regions of potential events based on the video features, which helps to reduce the ambiguity in the decoding process. For the relation prior, we design an event relation encoder that calculates the explicit pairwise relation between events and then encodes them into an attention mask. This attention mask serves as prior knowledge of

event relations, and is integrated into the self-attention operation of the decoder to facilitate better event interaction. With the incorporation of position and relation prior, the location distribution and semantic relations of multiple events in the video are explicitly clarified, which alleviates the training burden of the model, and improves both localization accuracy and caption coherence.

We evaluate our method on two popular dense video captioning datasets: ActivityNet Captions [5] and YouCook2 [18]. Our method achieves competitive performance against state-of-the-art methods. Ablation studies also demonstrate the effectiveness of the proposed position and relation prior. The main contributions of this paper can be summarized as follows:

- We design a novel dense video captioning framework, named PR-DETR, which injects position and relation prior to improve the localization accuracy and caption quality, simultaneously.
- We design a novel position-anchored query which injects the position prior to provide scene-specific search regions about potential events, improving the event localization accuracy.
- We design a novel event relation encoder which injects the relation prior to help model the event interactions in the decoder, improving the coherence of generated captions.

The rest of this paper is organized as follows: We briefly review the related work in Section 2. We present the technical details of our proposed method in Section 3. Then, extensive experiments and

analysis are presented in Section 4. Finally, we conclude the paper in Section 5.

2. Related works

2.1. Dense video captioning

Dense video captioning is a multi-task problem that combines event localization and event captioning to generate informative captions for long and untrimmed videos. Early approaches [14,19] typically employed a two-stage “localize-then-caption” framework, first localizing events in the video, and then captioning them. For example, Krishna et al. [5] first produce event proposals with a proposal module and then generate event captions with an LSTM. Wang et al. [19] propose a bidirectional proposal method to exploit both past and future contexts for temporal proposal generation. In these two-stage methods, the number of generated proposals is typically much larger than the actual number of events. As a result, these methods expand the prediction coverage of actual events, thereby improving recall performance.

However, the separation between event localization and captioning poses a gap in the model, limiting the two tasks to benefit each other. Additionally, the “localize-then-caption” strategy causes caption quality to depend on the quality of the proposals generated in the previous stage. To solve these problems and enrich inter-task interaction, recent studies propose to train the two sub-tasks jointly. To solve this problem and enrich inter-task interaction, recent studies [20–22] propose to train the two sub-tasks jointly. In particular, PDVC [7] defines dense video captioning as a set prediction task, and proposes an end-to-end framework to conduct event localization and captioning in parallel. CM2 [16], inspired by the philosophy of Retrieval-Augmented Generation (RAG) [23], introduces a new dense video captioning method with cross-modal retrieval from external memory. Vid2Seq [8] leverages unlabeled narrated videos and transcribed speech as multi-modal inputs for pretraining. It predicts a single sequence of discrete tokens that includes caption tokens and special time tokens. Zhou et al. [24] design a streaming model with a memory mechanism and a streaming decoding algorithm to process long videos and live video streams. E²DVC [25] proposes an event-equalized dense video captioning framework to treat all events equally. These end-to-end methods optimize both sub-task modules simultaneously, emphasizing the mutual enhancement between localization and captioning. This not only improves the model’s text generation capability but also increases the accuracy of event localization. Our method follows this end-to-end structure. Instead, we construct the position and relation prior information and incorporate them into the model to enhance both localization accuracy and caption coherence.

2.2. Transformer-based architecture

Transformer is a deep learning model architecture with attention mechanism for Natural Language Processing (NLP) and other sequence-to-sequence tasks. Dosovitskiy et al. [26] first adapt an encoder-only transformer for image recognition, yielding the vision transformer (ViT), which is widely used in other computer vision tasks [26–28]. Carion et al. [15] propose DETR, a transformer-based end-to-end object detector without hand-crafted components. PDVC [7] extends Deformable Transformer [29], a variant of DETR, into the dense video captioning task, which considers dense video captioning as a set prediction task to predict the locations and captions of multiple events in parallel. Based on PDVC, some other approaches [16,30] are proposed, these methods follow the general philosophy of detection transformer and implicitly learn the position and interaction of multiple events through a set of learnable queries and the attention mechanism. Instead, our method introduces the task-specific prior into the DETR-like architecture to reduce optimization difficulty and improve model performance.

2.3. Event proposal and relation modeling

The early two-stage methods usually generate a set of event proposals for event localization. Some of these methods [31,32] pre-define a vast number of anchors at different scales with regular intervals and utilize an evaluation network to score each anchor. Other works [33, 34] predict event boundaries and combine them to form event proposals. These methods all rely on a trainable network to conduct event proposal in a two-stage manner, making a multi-step training strategy necessary. Besides, they contain hand-crafted designs that require laborious tuning and preprocessing. What is different, our approach generates the position-anchored query which contains scene-specific position and semantic information about potential events to serve as proposals, realizing end-to-end proposal generation and training.

Modeling the relation among events is also crucial in the dense video captioning task. Early studies primarily adopt graph neural networks [35,36], where temporal event proposals are defined as nodes and their relations are represented as edges. During training, the graph network learns relation weights by passing messages across event nodes. Recent DETR-based methods [7,16] compute self-attention among queries to implicitly learn event relations from training data. Instead, we propose an event relation encoder to explicitly compute positional relations between events and encode them into a relation mask. The relation mask serves as prior information about event relations and is further integrated into the self-attention of the transformer decoder.

3. Method

3.1. Statistical analysis of event locations and semantics

In this section, we perform statistical analysis to demonstrate the distribution pattern of event locations and the relevance between event location and semantics. In Fig. 3(a) and (c), we visualize the ground-truth event distributions of two datasets. No matter which dataset, a distinct constraint can be observed in the distribution of event locations, that is, relatively long events tend to occur in the center of the video, which narrows down the search region for events.

To quantify the relevance between event location and semantics, we first design an IOU-style metric to measure the location correlation between the temporal boundaries of two events. Given two event boundaries (t_i^s, t_i^e) and (t_j^s, t_j^e) within a video, we compute the Location Correlation (LC) between them as follows:

$$LC = 1 - \frac{|\min(t_i^e, t_j^e) - \max(t_i^s, t_j^s)|}{\max(t_i^e, t_j^e) - \min(t_i^s, t_j^s)}, \quad (1)$$

where t_i^s and t_i^e denote the normalized start and end time of the i th event respectively. For the semantic similarity between two event descriptions, we adopt the BERT [37] model to extract features of event descriptions and calculate their cosine similarity. Fig. 3(b) illustrates the location correlation and semantic similarity of multiple events within a single video. As shown in the figure, there is a positive correlation between event locations and semantics, which means that two temporally connected events tend to have more relevant semantics. The above statistical results encourage us to explore and leverage the task-specific prior knowledge to help the model training for better localization and captioning.

3.2. Overview of PR-DETR

We illustrate the framework of our PR-DETR in Fig. 4. Given an untrimmed video $\mathcal{V}$ as input, our model outputs a set of event locations and corresponding captions $\{(t_n^s, t_n^e, S_n)\}_{n=1}^N$, where N denotes the number of events and S_n denotes the generated caption for n th event. For the input video $\mathcal{V}$, we first perform feature extraction and encoding (Section 3.3) to obtain the encoded video representations.

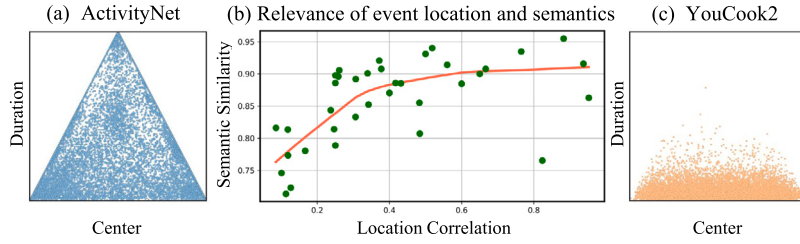


Fig. 3. Statistical analysis on dense video captioning datasets. (a) The ground-truth event distribution on ActivityNet. x-axis and y-axis denote the normalized event center and duration, respectively. (b) Statistical relevance between event location and semantics. The scatter points represent all event pairs within a single video, x-axis and y-axis represent their positional and semantic relationships, respectively. The red curve is the trend line. (c) The ground-truth event distribution on YouCook2.

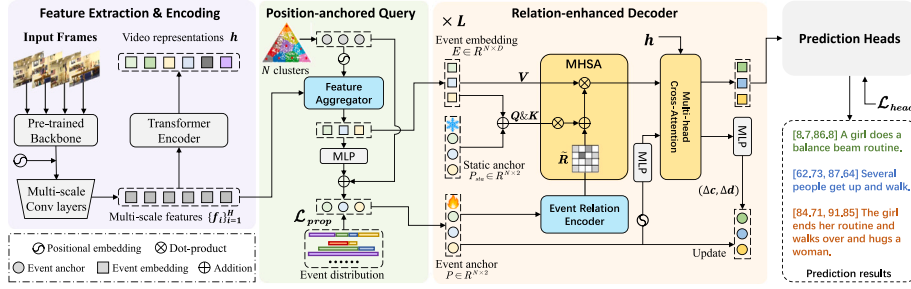


Fig. 4. Illustration of the proposed PR-DETR. We first obtain the video representations from the input video in feature extraction and encoding. Next, we construct and introduce the position and relation prior through the position-anchored query and relation-enhanced decoder. Finally, the output embedding of the decoder is fed into the prediction heads to produce the temporal boundaries and captions of dense events in the video.

Then, we design a set of position-anchored queries to probe the encoded video representations through the relation-enhanced decoder in dense event decoding (Section 3.4), producing the event-level features with rich semantic information. After that, we employ several parallel heads to predict the location and caption of events from the event-level features (Section 3.5). Finally, we present the training and inference of PR-DETR in Section 3.6.

3.3. Feature extraction and encoding

Given a video sequence $\mathcal{V}$, we first adopt a pre-trained backbone network CLIP [38] to extract the D -dimensional frame-level features at 1 FPS. Then, the frame-level features are rescaled to a fixed length V by interpolation for batch processing. To better perceive and predict events of varying durations in the video, we utilize H convolution layers to downsample the frame-level features to different temporal scales, obtaining the multi-scale features $\{\mathbf{f}_i\}_{i=1}^H$, where $\mathbf{f}_i \in \mathbb{R}^{V/2^{(i-1)} \times D}$. Finally, we apply the sinusoidal positional encoding [39] on the multi-scale features $\{\mathbf{f}_i\}_{i=1}^H$ and feed $\{\mathbf{f}_i\}_{i=1}^H$ into the transformer encoder along with their positional embedding to perform the inter-frame feature interaction at different temporal scales. The transformer encoder is designed with a stack of deformable transformer encoder layers, where each layer consists of a self-attention operation and a feedforward network. Through the transformer encoder, we obtain the encoded video representations $\mathbf{h} \in \mathbb{R}^{L \times D}$, where L represents the length of the flattened feature sequence.

3.4. Dense events decoding

In this section, we elaborate on the decoding process of our method, which includes how we construct the position and relation prior and integrate them into our framework.

Position-anchored Query. In the DETR-like methods, a set of event queries are used to probe the encoded features $\mathbf{h}$ for dense event prediction. Previous works initialize the event query with learnable embeddings. The design of the high-dimensional and scene-agnostic query makes the search region ambiguous and training difficult.

To alleviate this problem, we generate N position-anchored queries from the data distribution. It consists of two parts: high-dimensional event embedding $\mathbf{E} \in \mathbb{R}^{N \times D}$ and 2-dimensional event anchor $\mathbf{P} \in \mathbb{R}^{N \times 2}$. The event embedding serves as the carrier of event semantics, and the event anchor represents explicit locations of potential events $\mathbf{P} = \{(c_n, d_n)\}_{n=1}^N$, where c_n and d_n denote the center and duration of the n th query, respectively. We first obtain N centroids from the ground-truth data distribution through the K-means clustering algorithm, where N represents the number of queries. After that, we take the N initial event locations as anchors to aggregate the features of semantically related frames into event-level features through feature aggregation. The obtained event-level features are used as the event embedding. Then, we refine the initial N clustering centroids based on the event embedding to get the event anchors.

Specifically, we flatten the multi-scale features $\{\mathbf{f}_i\}_{i=1}^H$ and add them to their corresponding positional embeddings, obtaining $\mathbf{F} \in \mathbb{R}^{L \times D}$. To enable the model to better capture temporal position information, we separately project the centers and durations of N clustering centroids using sinusoidal positional encoding and then concatenate them to form the high-dimensional embedding as N event slots $\mathbf{S} \in \mathbb{R}^{N \times D}$. In the feature aggregator, we interact the event slots $\mathbf{S}$ with frame-wise features $\mathbf{F}$ for K iterations. In k th iteration, we first transform $\mathbf{S}^k$ and $\mathbf{F}$ as follows:

$$\tilde{\mathbf{S}}^k = \Phi(\text{LN}(\mathbf{S}^k)), \quad \tilde{\mathbf{F}} = \Phi(\text{LN}(\mathbf{F})), \quad (2)$$

where $\Phi(\cdot)$ represents a linear layer and $\text{LN}(\cdot)$ denotes layer normalization. Then, we calculate the attention weight matrix $\mathbf{A}_{\text{attn}}^k$ as follows:

$$\mathbf{A}_{\text{attn}}^k = \text{Softmax}\left(\frac{\tilde{\mathbf{S}}^k \tilde{\mathbf{F}}^T}{\sqrt{D}}\right), \quad (3)$$

where D is the hidden dimension of the model. We normalize the attention weight over the slot dimension to ensure the attention weights sum to one for each visual feature, obtaining $\tilde{\mathbf{A}}_{\text{attn}}^k$, where $\tilde{\mathbf{a}}_{m,n}^k =$

$\frac{a_{m,n}^k}{\sum_{m=1}^M a_{m,n}^k}$. The event slot is updated from S^k to S^{k+1} as follows:

$$\begin{aligned} \mathbf{U} &= S^k + \tilde{\mathbf{A}}_{attn}^k \Phi(\text{LN}(\mathbf{F})), \\ S^{k+1} &= \mathbf{U} + \text{MLP}(\text{LN}(\mathbf{U})). \end{aligned} \quad (4)$$

After K iterations, the event slot S^K contains preliminary semantic and location information of each potential event in the video. Therefore, we take S^K as the initial event embedding $\mathbf{E} \in \mathbb{R}^{N \times D}$ and utilize an MLP to project S^K into 2D-temporal offsets. Then we add the N clustering centers with the offsets to obtain the scene-specific event anchor $\mathbf{P} = \{(c_n, d_n)\}_{n=1}^N \in \mathbb{R}^{N \times 2}$. The event anchor and event embedding serve as the initial location and semantic search region of the potential events in the decoding process, respectively.

To supervise the generation of position-anchored queries, we introduce the proposal loss $\mathcal{L}_{prop}$. We first use the Hungarian algorithm [40] to find the optimal bipartite matching $\delta \in P_n$ between the predicted event anchor $\mathbf{P}$ and the ground-truth event location $\hat{\mathbf{P}}$ by minimizing the matching cost as follows:

$$\begin{aligned} \mathcal{L}_{giou}(\mathbf{P}_{\delta(i)}, \hat{\mathbf{P}}_i) &= 1 - \frac{|\mathbf{P}_{\delta(i)} \cap \hat{\mathbf{P}}_i|}{|\mathbf{P}_{\delta(i)} \cup \hat{\mathbf{P}}_i|}, \\ \hat{\delta} &= \arg \min_{\delta \in P_n} \mathcal{L}_{giou}(\mathbf{P}_{\delta(i)}, \hat{\mathbf{P}}_i), \end{aligned} \quad (5)$$

where $\mathcal{L}_{giou}$ is the generalized IOU loss [41] in its 1D temporal form. After obtaining the best matching $\hat{\delta}$ with the lowest matching cost, we use it as a new index of predictions to calculate the proposal loss as follows:

$$\mathcal{L}_{prop} = \sum_i^N \mathcal{L}_{giou}(\mathbf{P}_{\hat{\delta}(i)}, \hat{\mathbf{P}}_i). \quad (6)$$

Relation-enhanced Decoder. Although the position-anchored query provides the initial search region of events for the model, the relationship between events should also be considered to generate more coherent and reasonable captions. Previous methods implicitly learn the interaction between events from training data through the attention mechanism, increasing the optimization difficulty. Instead, we calculate and encode the relationship between events as an explicit prior and introduce it into self-attention for better event interactions in the relation-enhanced decoder.

Specifically, since the event anchor $\mathbf{P}$ contains the distribution pattern of the events, we design static anchors $\mathbf{P}_{sta} = \mathbf{P}$ to maintain the position guidance during decoding. In the relation-enhanced decoder, the static anchor $\mathbf{P}_{sta}$ maintains the position prior without updating, while the event anchor $\mathbf{P}$ is updated layer-by-layer for event localization. To match the input dimension of the self-attention, we project $\mathbf{P}_{sta}$ to D -dimensional space through the sinusoidal positional encoding and an MLP as follows:

$$\tilde{\mathbf{P}}_{sta} = \text{MLP}(\text{PE}(\mathbf{P}_{sta})), \quad (7)$$

where $\text{PE}(\cdot)$ denotes the sinusoidal position encoding.

From our observations and analysis, the semantic relations of events are correlated with their temporal positional relations. Therefore, we design the overlap-aware distance to capture the structural bias in event relations. The overlap-aware distance measures the positional relationship between events, which are subsequently encoded into a relation mask through sinusoidal positional encoding and a convolutional layer. This relation mask serves as the relation prior, guiding event interactions by modulating the self-attention weights. At l th decoder layer, we first calculate the pairwise overlap-aware distance between the N event anchors $\mathbf{P} = \{(c_n, d_n)\}_{n=1}^N$ to obtain relation matrix $\mathbf{R}$ as follows:

$$\begin{aligned} \beta &= |\min(r_i^e, r_j^e) - \max(r_i^s, r_j^s)|, \\ r(\mathbf{p}_i, \mathbf{p}_j) &= \left[\log \left(\frac{\beta}{d_i} + 1 \right), \log \left(\frac{d_i}{d_j} \right) \right], \end{aligned} \quad (8)$$

where r_i^s and r_i^e denote the normalized start and end time of the i th event, d_i is the duration of the i th event. $r(\mathbf{p}_i, \mathbf{p}_j)$ is an element of $\mathbf{R}$

and represents the relative positional relation between the i th and the j th events. The first term of $r(\mathbf{p}_i, \mathbf{p}_j)$ corresponds to the overlap-aware distance, and the second term indicates the relative duration of events. After obtaining the relation matrix $\mathbf{R}$, we adopt sinusoidal positional encoding to embed each element into a high-dimensional embedding. Then, a 1×1 convolution layer with normalization and activation function is used to perform the linear transformation on it as follows:

$$\tilde{\mathbf{R}} = \text{ReLU}(\text{LN}(\text{Conv}(\text{PE}(\mathbf{R})))) \in \mathbb{R}^{N \times N \times M}, \quad (9)$$

where $\tilde{\mathbf{R}}$ is the relation mask, M corresponds to the number of attention heads. Then, we integrate the relation mask $\tilde{\mathbf{R}}$ into self-attention. $\mathbf{Q}$, $\mathbf{K}$, and $\mathbf{V}$ are used to perform self-attention are calculated as follows:

$$\mathbf{Q} = \mathbf{W}_q(\mathbf{E}^l \oplus \tilde{\mathbf{P}}_{sta}), \mathbf{K} = \mathbf{W}_k(\mathbf{E}^l \oplus \tilde{\mathbf{P}}_{sta}), \mathbf{V} = \mathbf{W}_v \mathbf{E}^l, \quad (10)$$

where $\mathbf{W}_q$, $\mathbf{W}_k$, $\mathbf{W}_v$ are linear projection weights, and $\oplus$ denotes the element-wise addition operation. Based on $\mathbf{Q}$, $\mathbf{K}$ and $\tilde{\mathbf{R}}$, we compute the attention weight and obtain the updated event embedding $\tilde{\mathbf{E}}$ as follows:

$$\tilde{\mathbf{E}} = \text{Softmax} \left(\tilde{\mathbf{R}} + \frac{\mathbf{Q}\mathbf{K}^T}{\sqrt{D}} \right) \mathbf{V} + \mathbf{E}^l, \quad (11)$$

where D is the hidden dimension of the model. After self-attention, we take $\tilde{\mathbf{E}}$ as the query to interact with the video representations $\mathbf{h}$ through multi-head cross-attention (MHCA) and a feed-forward network (FFN), producing the updated event embedding $\mathbf{E}^{l+1}$.

3.5. Prediction heads

Following PDVC, we employ a parallel head structure, which includes a localization head, a captioning head, and an event counter to simultaneously predict N event boundaries, N captions, and event count from the event embedding of the final decoder layer $\mathbf{E}^L$.

Localization Head. The localization head is designed to predict event location and a confidence score for each event query. It employs an MLP to predict offsets (event center and duration) relative to the event anchor from the query embedding $\mathbf{E}^L$ output by the decoder. The confidence score indicates whether the predicted event belongs to the foreground. L_{cls} and L_{loc} are computed in the localization head. Specifically, L_{cls} is computed as the focal loss [42] between the predicted classification scores and the ground-truth labels, while L_{loc} is computed as the GIoU loss between the predicted temporal segments and the ground-truth segments.

Captioning Head. For the captioning head, we propose an LSTM with deformable soft attention to encourage the soft attention to focus more on the features around the event anchors. The captioning head takes attention-weighted features $\mathbf{z}_{n,t}$, query embedding $\mathbf{e}_n$, and the previous word $w_{n,t-1}$ as input to predict the probability of the next word $w_{n,t}$ based on the hidden state of LSTM. Finally, it generates the entire sentence $S_n = w_{n,1}, \dots, w_{n,M_j}$, where M_j represents sentence length. The captioning head produces $\mathcal{L}_{cap}$, which measures the cross-entropy between the predicted word probabilities and the ground-truth captions.

Event Counter. The event counter is proposed to predict the number of events in the video. It utilizes a max-pooling layer and a fully-connected layer to produce a high-dimensional vector $\mathbf{v}_{count}$ from query embedding $\mathbf{E}^L$, where each value in $\mathbf{v}_{count}$ represents the possibility of a specific number. The predict event number of the video N_{select} is determined by $N_{select} = \text{argmax}(\mathbf{v}_{count})$. The count loss $\mathcal{L}_{cnt}$ for this head is computed as the focal loss between $\mathbf{v}_{count}$ and the ground-truth event count.

We denote the loss function corresponding to the prediction heads as $\mathcal{L}_{head}$:

$$\mathcal{L}_{head} = \mathcal{L}_{cls} + \lambda_{loc} \mathcal{L}_{loc} + \lambda_{cnt} \mathcal{L}_{cnt} + \lambda_{cap} \mathcal{L}_{cap}, \quad (12)$$

where λ_{loc} , λ_{cnt} , and λ_{cap} represent the balancing coefficients for the corresponding losses.

Table 1

Dense captioning performance comparison with the state-of-the-art methods. B4, M, C, S denote BLEU4, METEOR, CIDEr and SODA_c, respectively. The best results are highlighted in bold, and the second-best are underlined.

Method	Venue	Features	YouCook2				ActivityNet			
			B4↑	M↑	C↑	S↑	B4↑	M↑	C↑	S↑
Video LLMs:										
Video-LLaMA [43]	EMNLP'23	–	–	–	6.10	1.80	–	–	–	–
VideoChat [44]	SCIS'23	–	–	–	8.10	2.20	–	–	–	–
TimeChat [45]	CVPR'24	–	–	–	10.30	3.10	–	–	–	–
Pre-training:										
Streaming GIT [24]	CVPR'24	CLIP	–	3.60	15.40	3.20	–	9.00	41.20	6.60
Vid2Seq [8]	CVPR'23	CLIP	–	9.30	47.10	7.90	–	8.50	30.10	5.80
ECHR [46]	TCSVT'20	C3D	–	3.82	–	–	1.29	7.19	14.71	3.22
MT [47]	CVPR'18	TSN	0.30	3.18	6.10	–	1.15	4.98	9.25	–
PDVC [7]	ICCV'21	TSN	0.80	4.74	22.71	4.42	1.78	7.96	28.96	5.44
PDVC ^a [7]	ICCV'21	CLIP	1.31	5.47	29.20	4.95	2.11	8.09	29.68	5.87
CM2 [16]	CVPR'24	CLIP	1.63	6.08	31.66	5.34	2.38	8.55	33.01	6.18
E ² DVC [25]	CVPR'25	CLIP	1.68	6.11	34.26	5.39	2.43	8.57	33.63	6.13
PR-DETR	Ours	CLIP	1.89	6.48	37.30	5.43	2.58	8.72	<u>33.16</u>	<u>6.13</u>

^a denotes results reproduced from official implementation using CLIP features.

3.6. Training and inference

Our method is trained end-to-end, during training, PR-DETR outputs a set of N event locations and corresponding captions $\{(t_n^s, t_n^e, S_n)\}_{n=1}^N$. We adopt the Hungarian algorithm [40] to match the predicted N events with ground truths, obtaining the best permutation. Then, we use the best permutation as a new index of predictions to calculate the total training loss $\mathcal{L}$ as follows:

$$\mathcal{L} = \mathcal{L}_{head} + \lambda_{prop} \mathcal{L}_{prop}, \quad (13)$$

where $\mathcal{L}_{prop}$ represents the proposal loss used to supervise the generation of position-anchored queries in Section 3.4, and λ_{prop} denotes the weight. During inference, we select top- N_{select} detected events as final results $\{(t_n^s, t_n^e, S_n)\}_{n=1}^{N_{select}}$ based on the confidence of each query.

4. Experiment

4.1. Experimental settings

Datasets. We evaluate PR-DETR on the two popular dense video captioning datasets: ActivityNet Captions and YouCook2. ActivityNet Captions contains 20k untrimmed videos of various human activities. On average, each video spans 120s and is annotated with 3.7 temporally localized sentences. We follow the standard data split for training, validation, and testing. YouCook2 consists of 2k untrimmed cooking procedure videos, with an average duration of 320 s per video and 7.7 temporally localized sentences per annotation. This dataset focuses on cooking scenes, and most videos contain one main actor cooking while describing the recipe. Since our method uses CLIP features extracted from raw videos, we only use those videos that are currently accessible on YouTube as training data, which is approximately 7% less than the original video count.

Evaluation Metrics. We evaluate our method in terms of event localization and caption generation, respectively. For dense captioning performance, we use the official evaluation tool of ActivityNet Challenge, which calculates matched pairs between generated events and the ground truth across IoU thresholds of {0.3, 0.5, 0.7, 0.9} and computes captioning metrics (BLEU4 [48], METEOR [49], CIDEr [50]) over the matched pairs. Moreover, we incorporate SODA_c [51], which considers all event captions from the same video. For localization performance, we report the average precision and average recall across IoU thresholds of {0.3, 0.5, 0.7, 0.9}, along with the overall F1 score.

Implementation Details. We use pre-trained CLIP ViT-L/14 to extract video features at 1 FPS. We set the number of position-anchored queries to 10 for ActivityNet Captions and 100 for YouCook2, and the iteration K of slot attention is set to 3 in both datasets. The hidden

dimension D of our model is set to 512, and the number of convolution layers H is set to 4. In the relation-enhanced decoder, the dimension of relation embedding is chosen as 16, and the number of attention heads M is set to 8. Our model is trained end-to-end. For both datasets, we employ the Adam optimizer with an initial learning rate of 5×10^{-5} and 30 training epochs, the batch size is set to 1. The hyper-parameters for weight scaling λ_{loc} , λ_{cnt} , λ_{cap} , and λ_{prop} are set to 2, 1, 1, and 1, respectively.

4.2. Comparison with state-of-the-art methods

In this section, we compare PR-DETR with previous methods in terms of dense captioning and event localization on the YouCook2 and ActivityNet Captions datasets. We utilize only visual features as inputs, excluding extra modalities like transcribed speech or audio. Additionally, our approach is trained purely on the specific DVC dataset without leveraging any additional videos.

Dense Captioning Performance. In Table 1, we show the dense captioning performance of our method and various dense video captioning approaches on YouCook2 and ActivityNet Captions datasets. Our method achieves competitive performance on both datasets, outperforming its counterparts PDVC and CM2 in almost all metrics. By injecting prior knowledge into our model, we reduce the demand for data, achieving comparable performance on less training data compared with Vid2seq and streaming GIT, which use an extra 1 million videos on YouCook2 and 15 million videos on ActivityNet for pertaining. This is because the proposed event relation encoder emphasizes the relationship between events with co-occurring semantics, maintaining coherence between event captions, thus improving caption quality. Table 1 shows the performance of several Video Large Language Models (VLLMs) after fine-tuning on the DVC dataset, as reported in [45]. Notably, our method significantly outperforms existing VLLMs (e.g., TimeChat), indicating the value of exploring task-specific priors for downstream applications.

Localization Performance. We also compare the localization performance of our method with other approaches on both datasets. As shown in Table 2, our PR-DETR achieves the best precision and F1 scores and second recall scores in YouCook2 dataset. In ActivityNet Captions, our method achieves the best precision and recall scores, leading to the highest F1 score. This is mainly because we initialize scene-specific queries from ground-truth event distribution, which eliminates some implausible event locations and provides a more reliable initial event search region, thus improving the model's localization ability. In contrast, existing VLLMs and pretraining methods focus more on the model's captioning ability, which helps less with event localization.

Table 2

Event localization performance comparison with the state-of-the-art methods. The best results are highlighted in bold, and the second-best are underlined.

Method	Venue	YouCook2			ActivityNet		
		Pre.↑	Rec.↑	F1 ↑	Pre.↑	Rec.↑	F1 ↑
Video LLMs:							
Video-LLaMA [43]	EMNLP'23	–	–	14.20	–	–	–
TimeChat [45]	CVPR'24	–	–	19.50	–	–	–
Pre-training:							
Streaming GIT [24]	CVPR'24	–	–	16.60	–	–	50.90
Vid2Seq [8]	CVPR'23	27.80	27.90	27.30	53.90	52.70	52.40
MFT [52]	ECCV'18	–	–	–	51.41	24.31	33.01
PDVC ^a [7]	ICCV'21	32.58	23.48	27.29	56.72	53.22	54.91
CM2 [16]	CVPR'24	33.38	24.76	28.43	56.81	53.71	55.21
E ² DVC [25]	CVPR'25	34.75	24.36	28.64	57.70	54.67	56.14
PR-DETR	Ours	36.11	24.06	28.87	57.48	53.82	55.59

^a denotes results reproduced from official implementation using CLIP features.

Table 3

Ablation studies on position and relation prior of PR-DETR on YouCook2 validation set. P denotes the position-anchored query and R denotes the relation-enhanced decoder.

Method	B4↑	M↑	C↑	S↑	Pre.↑	Rec.↑
Base	1.31	5.47	29.20	4.95	32.58	23.48
Base+P	1.39	5.95	31.54	5.40	34.34	24.36
Base+R	1.68	6.13	33.44	5.16	34.76	22.94
Base+P+R (Ours)	1.89	6.48	37.30	5.43	36.11	24.06

Table 4

Effect of the iteration number K in feature aggregator on YouCook2 validation set.

Iteration K	B4↑	M↑	C↑	S↑	Pre.↑	Rec.↑
1	1.79	6.69	36.29	5.14	37.36	22.23
2	1.78	6.41	36.09	5.41	35.89	24.47
3	1.89	6.48	37.30	5.43	36.11	24.06
4	1.77	6.64	35.56	5.29	36.86	22.81

4.3. Ablation studies

In this section, we conduct ablation studies to investigate the impact of key components of PR-DETR.

Effect of Position and Relation Prior. We first conduct ablation experiments to verify the effectiveness of the position prior and relation prior introduced in the position-anchored query and relation-enhanced decoder, respectively. As shown in Table 3, the introduction of both position and relation prior can improve all metrics effectively. The position prior improves the localization performance, while the relation prior mainly increases the captioning performance. This is because the position-anchored query with position prior provides more reliable event search regions, leading to higher precision and recall scores. The improvement in captioning performance validates the effectiveness of modeling explicit event relations for better caption quality and coherence. Combining the position-anchored query and relation-enhanced decoder, our method achieves significant performance improvement in both localization and captioning sub-tasks.

Effect of Iteration Number K in Feature Aggregator. We investigate the effect of the iteration number K in the feature aggregator. Table 4 shows the performance corresponding to the varying iteration number K . The iteration number K is related to the granularity of feature aggregation. As K increases, the slot embedding groups frame-wise features in a wider temporal range, which gradually improves the model performance. A large number of K makes the model more likely to ignore shorter events, which leads to performance degradation, especially on YouCook2 dataset with many short events. Consequently, the iteration number K is determined to be 3 for the best performance.

Component-wise Analysis of Position-anchored Query. We conduct component-wise ablative experiments on the YouCook2 dataset

Table 5

Component-wise ablation results of the position-anchored query on YouCook2 validation set. EQ means initializing explicit queries with the clustering centers, FA means feature aggregator and SA denotes static anchor P_{sta} .

EQ	FA	SA	B4↑	M↑	C↑	S↑	Pre.↑	Rec.↑
×	×	×	1.68	6.13	33.44	5.16	34.76	22.94
✓	×	×	1.75	6.23	34.40	5.25	34.76	22.81
×	✓	×	1.74	6.16	35.03	5.25	34.78	23.02
✓	×	✓	1.67	6.16	34.86	5.36	35.04	23.56
✓	✓	×	1.84	6.45	36.72	5.36	36.19	23.83
✓	✓	✓	1.89	6.48	37.30	5.43	36.11	24.06

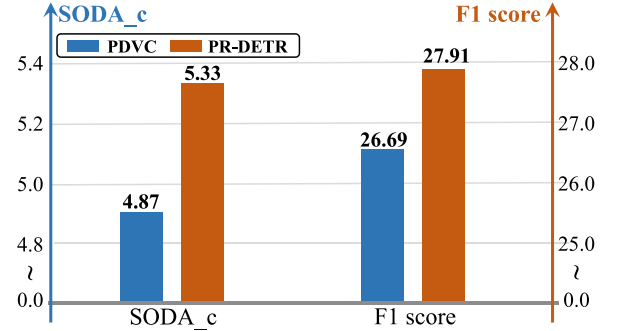


Fig. 5. Performance comparison between PDVC and our PR-DETR on the ActivityNet-YouCook2 mixed dataset.

to further analyze the effectiveness of each component within the position-anchored query. As shown in Table 5, initializing explicit queries with clustering centroids effectively improves the model performance (Row 2). Note that performing feature aggregation without initialized explicit queries only brings a slight performance gain due to the lack of guidance from the clustered event locations (Row 3). When we incorporate both explicit query and feature aggregator into our method, the model performance is improved by a considerable margin (Row 5), which demonstrates the significance of generating scene-specific queries with the position prior. We also take the above scene-specific event anchor as the static anchor for the relation-enhanced decoder to maintain the position prior during decoding. This design further enhances the performance (compare Row 5 and Row 6), leading to the best result.

Effect of the Proposal Loss. The balancing parameter λ_{prop} controls the impact of proposal loss $\mathcal{L}_{prop}$, which is designed to supervise the query generation. To study the effect of $\mathcal{L}_{prop}$, we report the performance of different λ_{prop} in Table 6. The result shows that the best performance is obtained when $\lambda_{prop} = 1$, too large or too small λ_{prop} leads to performance degradation. When λ_{prop} is too small, the model struggles to learn scene-specific representations from the position priors, whereas when λ_{prop} is too large, it can interfere with the primary

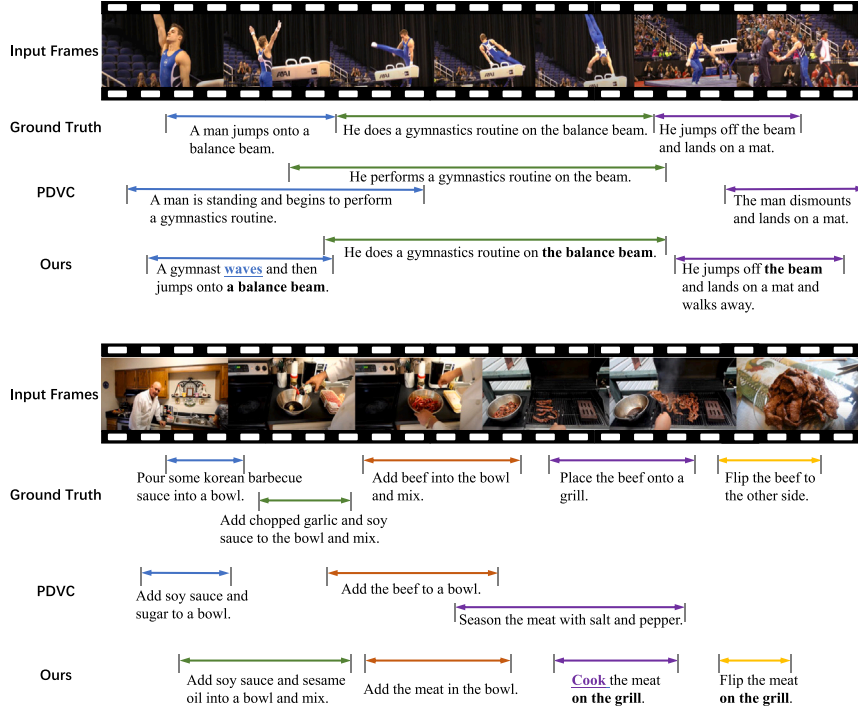


Fig. 6. Visualization of dense video captioning results of PDVC and our PR-DETR on ActivityNet Captions and YouCook2 validation set. Lines in different colors represent the temporal location of different events. The co-occurring semantics and fine-grained actions recognized by our method are highlighted.

Table 6

Performance of different balancing parameter λ_{prop} on YouCook2 validation set. λ_{prop} controls the impact of proposal loss $\mathcal{L}_{prop}$.

λ_{prop}	B4↑	M↑	C↑	S↑	Pre.↑	Rec.↑
0.5	1.75	6.41	35.47	5.10	34.82	22.31
1	1.89	6.48	37.30	5.43	36.11	24.06
2	1.77	6.59	36.18	5.14	37.13	22.67
3	1.78	6.56	36.12	5.23	36.78	23.18

Table 7

Performance of different relation encoding methods in the event relation encoder on the YouCook2. ‘No Relation Encoder’ represents the baseline method without the relation prior, ‘Center-aware Encoding’ means calculating the relative distance between event centers, and ‘Overlap-aware Encoding’ denotes calculating the proposed overlap-aware distance.

Methods	B4↑	M↑	C↑	S↑	Pre.↑	Rec.↑
No Relation Encoder	1.39	5.95	31.54	5.40	34.34	24.36
Center-aware Encoding	1.72	6.13	35.25	5.30	34.23	22.28
Overlap-aware Encoding	1.89	6.48	37.30	5.43	36.11	24.06

localization and captioning losses. Experimental results validate the impact of the proposal loss. Consequently, we choose λ_{prop} to be 1 in our paper for the best performance.

Comparison of Alternative Relation Encoding Methods. To obtain the relation mask, we design two approaches to compute pairwise relative distance between event boundaries, named center-aware distance and overlap-aware distance, respectively. The center-aware distance calculates the relative distance between event centers. The overlap-aware encoding follows the IOU-like metrics to indicate whether two event boundaries are temporally connected, as described in Eq. (8). For two temporally connected events, the overlap-aware distance yields the smallest distance. As shown in Table 7, the overlap-aware distance achieves better performance compared to the center-aware distance

on both captioning and localization metrics. This is because overlap-aware distance takes into account the overlap and duration of events, which can more accurately depict the positional relationship between two events compared to center-aware distance. The results demonstrate the value of exploring more accurate metrics of event location relationships.

Results on the Mixed Dataset. We mix ActivityNet and YouCook2 datasets to construct a mixed dataset with more diverse and complex patterns, which is possible in real-world scenarios. Specifically, we sample 10% of the training videos from ActivityNet and combine them with the training videos from YouCook2 dataset. For the evaluation, we mix the validation sets of the two datasets in the same ratio. Afterwards, we apply the training and evaluation settings for ActivityNet to conduct experiments on the mixed dataset. As shown in Fig. 5, our method achieves substantial improvements over PDVC. By incorporating explicit prior knowledge, our method is more effective in learning complex event patterns of the input data.

4.4. Visualization

We visualize the prediction results of different models in Fig. 6. Benefiting from the introduction of the position prior, our method predicts more accurate event boundaries. Moreover, the explicit relation we construct from event locations helps to model the interaction between events, which improves the coherence of dense captions in the video. As a result, the captions generated by our method contain the co-occurring target in the context, e.g., ‘the balance beam’. Our model also successfully recognizes some fine-grained actions that are not mentioned in the ground-truth, such as ‘waves’ and ‘cook’.

4.5. Analysis of failure case

We also present some failure cases of our method to offer a more comprehensive demonstration. As shown in Fig. 7, PR-DETR produces predictions different from the ground truth when dealing with scenarios

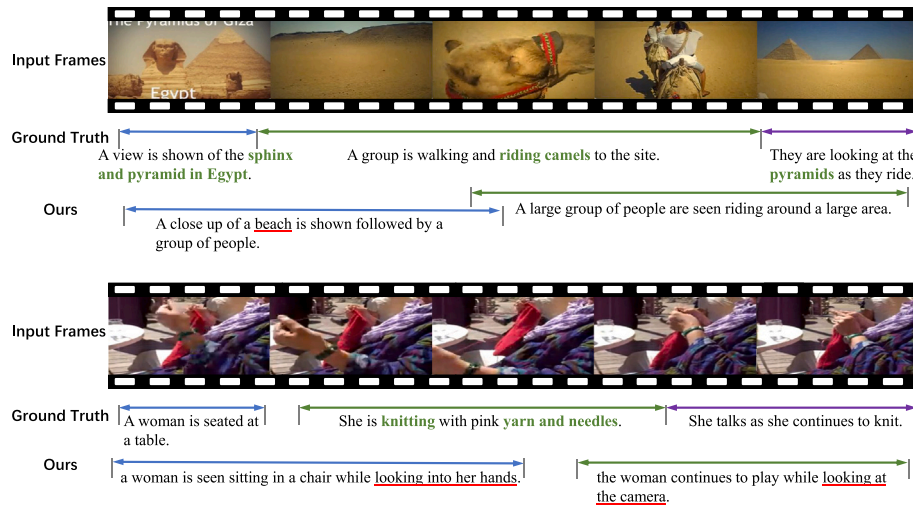


Fig. 7. Some failure cases of PR-DETR. Semantics missed by our method are highlighted in green. Incorrect phrases are underlined in red.

containing uncommon objects and fine-grained actions. For example, in the first scenario, our method incorrectly identifies the desert as a beach and fails to recognize the less common objects, the sphinx and the pyramid in the video. The second scene shows a woman sitting and knitting without scene changes or camera movements. In this case, PR-DETR fails to recognize the concept of knitting and instead describes it as a woman looking into her hands. Incorporating a stronger action recognition model or concept-aware knowledge could enhance the model's ability to recognize rare objects and fine-grained actions, thereby improving DVC performance. This is also a promising direction for our future work.

5. Conclusion

In this paper, we propose PR-DETR, which injects the explicit position and relation prior knowledge into the detection transformer to help the model training for better event localization and caption generation. The position-anchored query provides more reliable scene-specific position and semantic information about potential events. The event relation encoder constructs the relation mask between event queries as relation prior to guide the event interaction in the relation-enhanced decoder, improving the semantic coherence of the captions. Our PR-DETR offers an innovative solution for leveraging task-specific knowledge in dense video captioning. Ablation studies validate the effectiveness of the position and relation prior. Experiment results on ActivityNet Captions and YouCook2 datasets also demonstrate the improved performance of PR-DETR.

CRediT authorship contribution statement

Yizhe Li: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Sanping Zhou:** Supervision, Resources, Project administration, Investigation, Funding acquisition. **Zheng Qin:** Writing – review & editing, Methodology, Formal analysis. **Le Wang:** Supervision, Resources, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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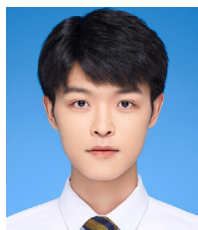
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